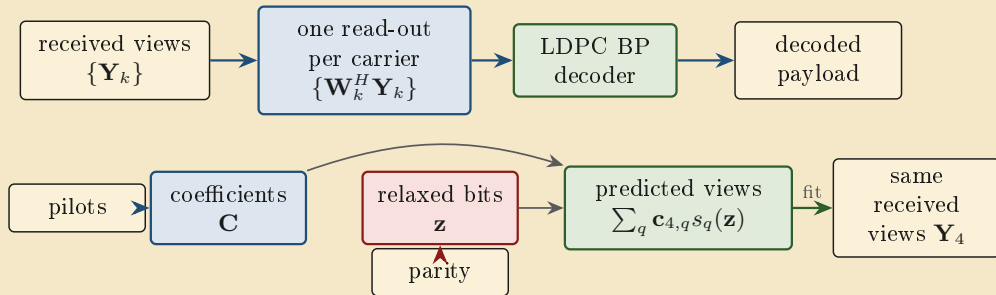


# The Joint Demodulation and Decoding Problem

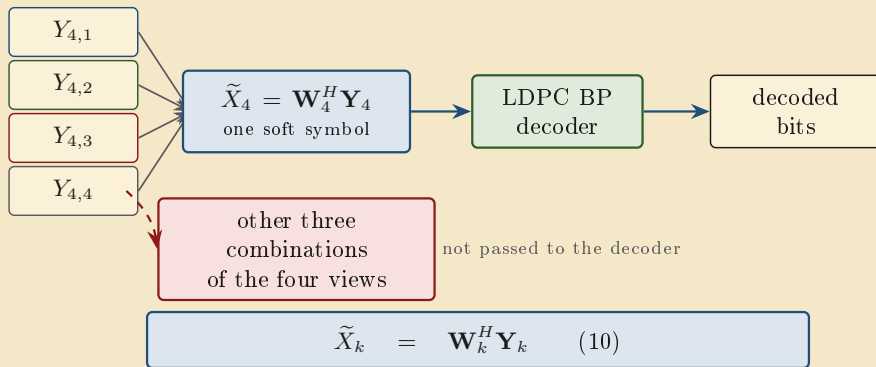
Carrier 4 inside the frame-wide receiver



The combiner is a read-out. The joint problem scores the coefficients and relaxed bits against the received views.

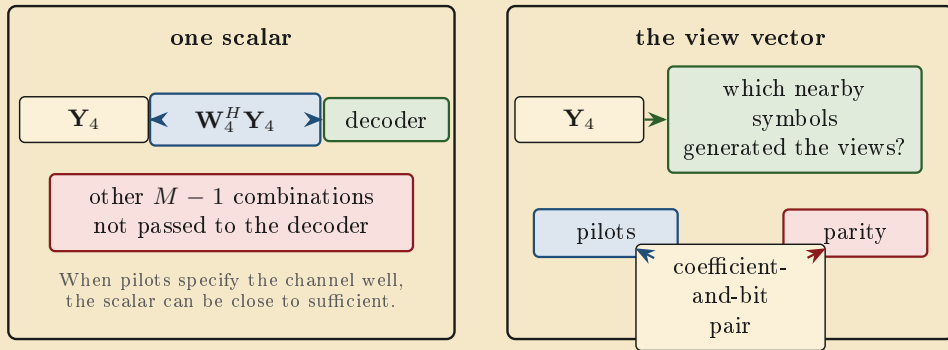
# The read-out

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*The read-out keeps one combination of a carrier's views for the decoder. Here  $\mathbf{Y}_k$  contains the  $M$  views and  $\mathbf{W}_k$  contains one weight per view.*

# Scalar or vector?



*The joint search retains the view vector, uses pilots to constrain the coefficients, and uses parity checks to restrict the coded bits.*

# A worked example

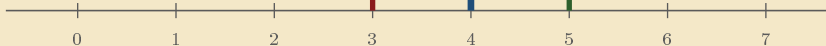
$N = 8$  carriers,  $M = 4$  views  
common carrier-frequency offset 0.20 bins

received bin  $k = 4$   
wanted carrier  $q = 4$ , neighbouring carrier  $q = 5$

$$|H_{4,4}| = 0.9365$$

$$|H_{4,3}| = 0.2378$$

$$|H_{4,5}| = 0.1618$$



With zero offset,  $\mathbf{H} = \mathbf{I}$ .

*The following pages keep carrier 4, its four views, and neighbouring carrier 5 visible. This is a toy calculation, not a measured Red channel.*

# Four views

---

| $m$ | $c_{4,4,m}$        | $c_{4,5,m}$         |
|-----|--------------------|---------------------|
| 1   | $0.2485 + 0.0196j$ | $0.1985 + 0.1011j$  |
| 2   | $0.2303 + 0.0954j$ | $-0.1575 + 0.1575j$ |
| 3   | $0.1895 + 0.1619j$ | $-0.1011 - 0.1985j$ |
| 4   | $0.1302 + 0.2125j$ | $0.2200 - 0.0348j$  |
| sum | $0.7985 + 0.4893j$ | $0.1598 + 0.0253j$  |

$$\sum_{m=1}^M c_{k,q,m} = H_{k,q} \quad (29)$$

when  $\sum_{m=1}^M \mathbf{S}_m = \mathbf{I}$

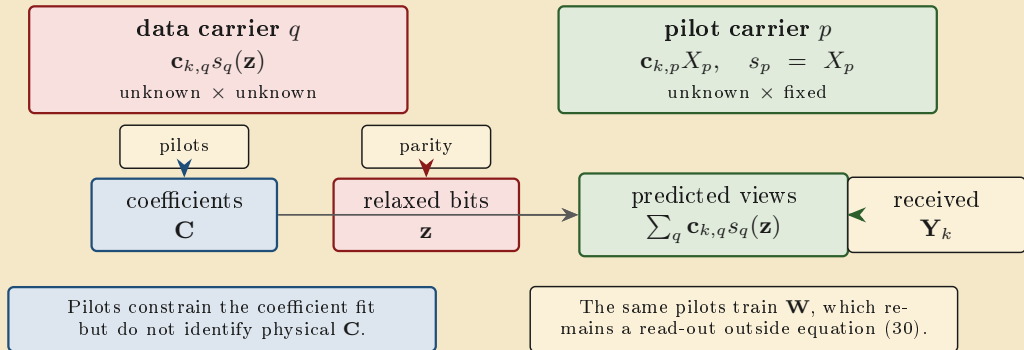
maximum difference over all pairs  
 $2.2 \times 10^{-16}$

wanted components add nearly in phase;  
 neighbour components nearly cancel

windowed FFT:  $Y_{4,1} = -0.1396 + 0.3878j$  complete eight-  
 carrier coefficient mix:  $\sum_{q=0}^7 c_{4,q,1} X_q = -0.1396 + 0.3878j$   
 difference  $1.1 \times 10^{-16}$

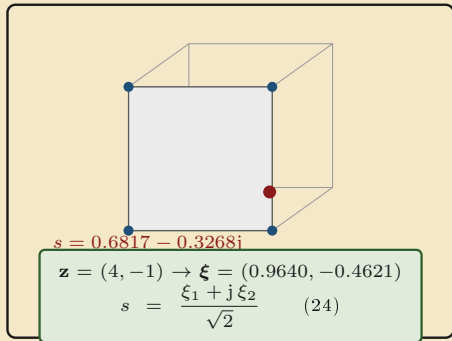
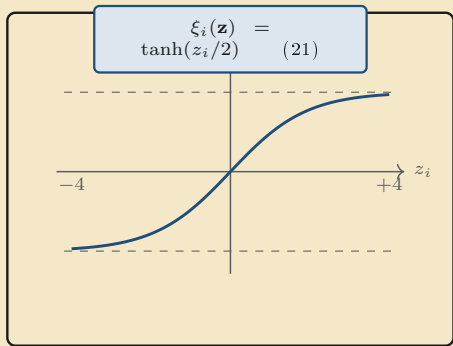
*Each coefficient is one time-local part of the same channel coefficient. The receiver retains nearby pairs in equation (28); omitted pairs are not asserted to be physically zero.*

# Pilots



*Pilots constrain the coefficients, while parity checks restrict the coded bits. Both affect the predicted views through different variables.*

# Relaxed symbol



*Hard bits give the four quadrature phase-shift keying points. Relaxed bits move the symbol inside the square and provide a differentiable path. The square is one face of the cube the coded bits span.*

# Parity penalty

$$\mathcal{R}_{\text{par}}(\mathbf{z}) = \sum_j \left( 1 - \prod_{i \in \mathcal{N}_j} \xi_i(\mathbf{z}) \right)^2 \quad (22)$$

where  $\mathcal{N}_j$  is the set of bits in parity check  $j$ .

**hard bits**  
even parity  
product +1.0000  
penalty 0.0000



**relaxed values**  
correct signs  
product +0.4032  
penalty 0.3561



**one sign changed**  
wrong parity  
product -0.4032  
penalty 1.9691



*The penalty vanishes on codewords. Relaxed values with the correct signs remain costly until their magnitudes approach one.*



# One objective

$$\mathcal{J}(\mathbf{C}, \mathbf{z}) = \underbrace{\lambda_d \sum_{k \in \mathcal{K}} \left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} s_q(\mathbf{z}) \right\|_2^2}_{\text{signal term}} + \underbrace{\gamma_C \|\mathbf{C}\|_F^2}_{\text{coefficient prior}} + \underbrace{\lambda_p \mathcal{R}_{\text{par}}(\mathbf{z})}_{\text{parity penalty}} \quad (30)$$

where  $\lambda_d$ ,  $\gamma_C$ , and  $\lambda_p$  weigh the view fit, coefficient energy, and parity penalty. The view noise and coefficient prior are isotropic complex Gaussian.

## signal check

true coefficient-and-bit pair

noise  $\sigma = 0.05$

**0.0185**

## one parity check

hard even: 0

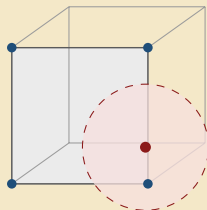
relaxed signs: 0.3561

one wrong sign: 1.9691

These are separate checks. They do not form a complete numerical  $\mathcal{J}$  and do not share a scale.

*Equation (30) scores  $\mathbf{C}$  and  $\mathbf{z}$  against the received views. The combiner  $\mathbf{W}$  remains outside the objective.*

# The ring



the same symbol  $s = 0.6817 - 0.3268j$ ,  $v = 0.4285$

$$\begin{aligned}\mu_q &= s_q(\mathbf{z}) \\ v_q &= 1 - |\mu_q|^2\end{aligned}$$

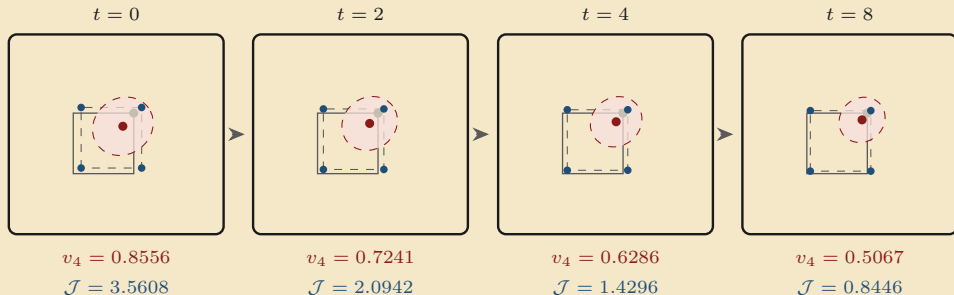
pilots keep their  
known symbols:

$$\mu_q = X_q, v_q = 0, \text{ no ring}$$

a confident symbol shrinks the ring;  
an uncertain one grows it

*The ring has radius  $\sqrt{v_q}$  around the symbol mean. The errors are zero-mean and mutually uncorrelated, so it is a circle, not an ellipse.*

# Gradient steps



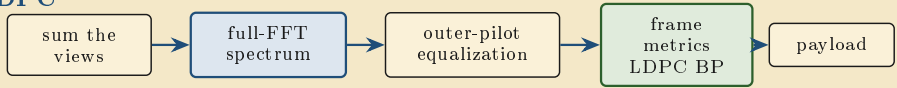
carrier 4 climbs toward its corner  $(+1 + j)/\sqrt{2}$ ; the mean  $v$  over all eight carriers falls from 0.8556 to 0.4816

*Each gradient step lowers the objective, and the signal likelihood rises as the objective falls. Here the coefficients are held at their true values and plain gradient steps with a fixed step size move only  $\mathbf{z}$ ; the receiver instead solves  $\mathbf{C}$  conditionally and moves  $\mathbf{z}$  with Adam.*

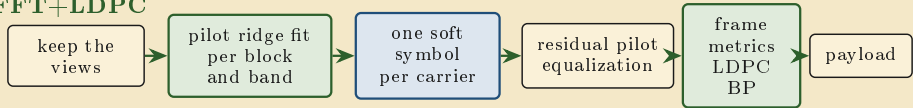
# Receiver algorithms

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## OFDM+LDPC



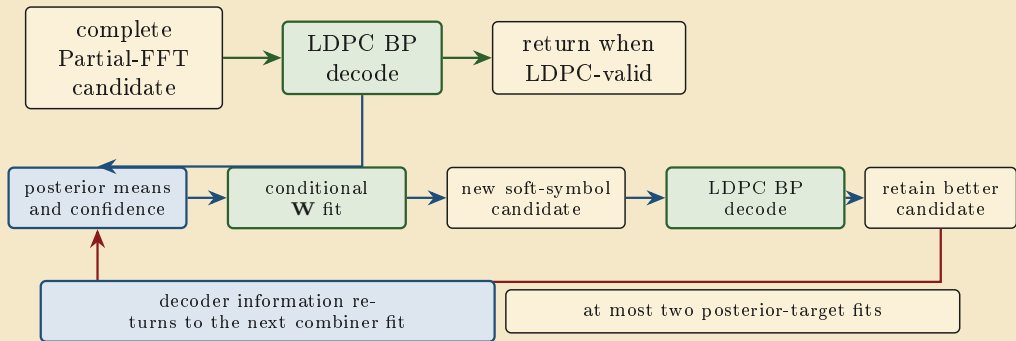
## Partial-FFT+LDPC



Neither baseline returns decoder output to another receiver update.

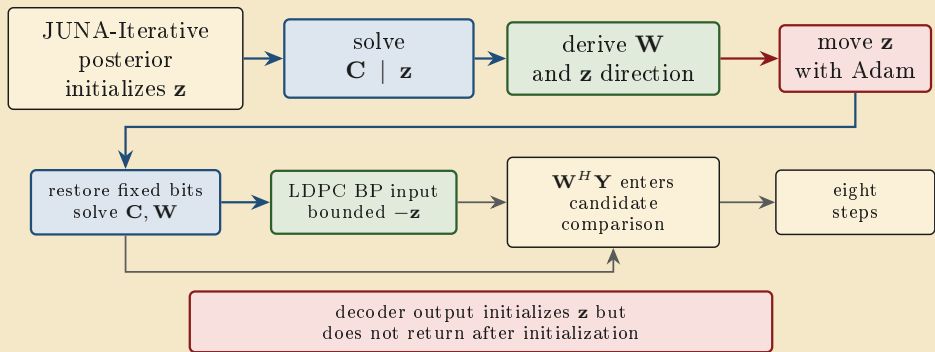
# JUNA-Iterative

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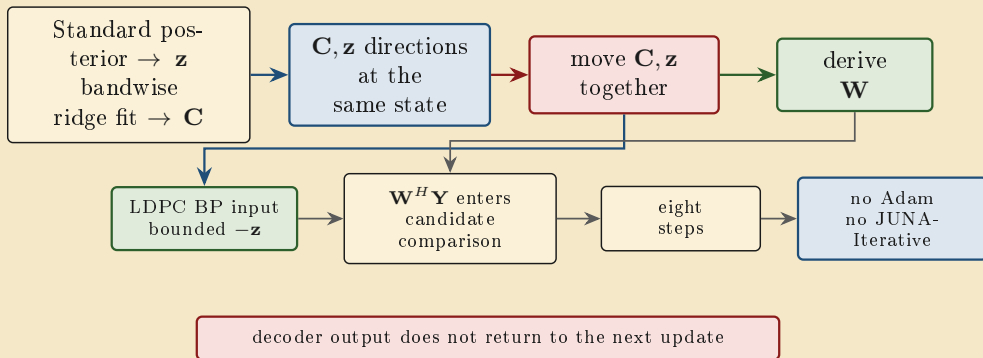
*Like turbo equalization, JUNA-Iterative repeats equalization and decoding. It forms no  $\mathbf{C}$ , has no free  $\mathbf{z}$ , and never evaluates equation (30).*

# JUNA- $C, z$



*JUNA- $C, z$  conditionally solves the coefficients and moves only the relaxed bits. Belief propagation receives  $-\mathbf{z}$ , not  $\mathbf{W}^H \mathbf{Y}$ .*

# JUNA-Direct- $C, z$



*JUNA-Direct- $C, z$  moves the coefficient and relaxed-bit variables together. The weights are derived from the updated coefficients and remain outside equation (30).*

# Takeaway

|                                                    |                                                                                |                   |
|----------------------------------------------------|--------------------------------------------------------------------------------|-------------------|
| <b>OFDM+LDPC</b><br>sum views                      | outer-pilot equalization $\rightarrow$ LDPC BP                                 | payload           |
| <b>Partial-FFT+LDPC</b><br>keep views              | pilot-trained $\mathbf{W}$ $\rightarrow$ one soft symbol $\rightarrow$ LDPC BP | payload           |
| <b>JUNA-Iterative</b><br>keep views                | decoder means return $\rightarrow$ conditional $\mathbf{W}$ fit                | payload           |
| <b>JUNA-<math>C, z</math></b><br>keep views        | solve $\mathbf{C} \mid \mathbf{z} \rightarrow$ move $\mathbf{z}$               | no decoder return |
| <b>JUNA-Direct-<math>C, z</math></b><br>keep views | move $\mathbf{C}, \mathbf{z}$ together $\rightarrow$ derive $\mathbf{W}$       | no decoder return |

Pilots constrain  $\mathbf{C}$ ; parity checks constrain  $\mathbf{z}$ ; equation (30) scores the coefficient-and-bit pair in the received views.

*Only JUNA-Iterative returns decoder information to the next update.*



# Posterior variance

worked symbol

$$\mu = s = 0.6817 - 0.3268j$$

$$|\mu|^2 = 0.5715$$

symbol variance

$$v = 1 - |\mu|^2$$

$$\mathbf{v} = \mathbf{0.4285}$$

$$\overline{\mathcal{J}}_{\text{sig}} = \lambda_d \sum_{k \in \mathcal{K}} \left[ \left\| \mathbf{Y}_k - \sum_{q \in \mathcal{B}_K(k)} \mathbf{c}_{k,q} \mu_q \right\|_2^2 + \sum_{q \in \mathcal{B}_K(k)} v_q \|\mathbf{c}_{k,q}\|_2^2 \right] \quad (68)$$

where  $\mu_q$  is the symbol mean and  $v_q$  is its variance. Symbol errors are zero-mean and mutually uncorrelated; pilots and fixed carriers use  $v_q = 0$ .

This alternative is not used for the reported mean-only  $C, z$  results. It is not a marginalized likelihood and omits parity-induced correlations.

*Equation (68) replaces only the signal term of equation (30).*